

Akshat Raj

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TECHNICAL SKILLS

Languages & Databases: Python (Pandas, NumPy, Scikit-Learn), TypeScript, JavaScript (Node.js), SQL (MySQL, PostgreSQL, T-SQL), MongoDB, ChromaDB, Qdrant (Vector DB)

Data Analytics & BI: Power BI (DAX, Power Query), Tableau, Excel (Power Query, Pivot Tables), ETL Pipelines, KPI Development

AI & LLM Engineering: LangChain.js, LangGraph, GPT-4o/Llama API, Multi-Agent Systems, RAG, Hybrid Search (BM25/Dense), Rerankers, Ragas, LangSmith

Backend, Cloud & DevTools: FastAPI, Express.js, REST APIs, AWS (S3, EC2, Lambda, RDS, IAM), Docker, GitHub Actions, Git, Postman

Core Competencies: Cloud Architecture, Data Warehousing, Medallion Architecture, Dimensional Modeling (Star Schema), EDA, A/B Testing

EXPERIENCE

Data Analyst & ML Intern

CESC Rajasthan, RP-Sanjiv Goenka Group

May 2026 – Present

Rajasthan, India

- Engineered and evaluated a short-term load forecasting model using Python (Scikit-Learn, XGBoost) to predict hourly substation electricity demand, achieving a **96%** forecasting accuracy rate.
- Processed raw time-series consumption data from **10K+** smart meters using Pandas and NumPy, conducting Exploratory Data Analysis (EDA) to detect grid anomalies and mitigate potential revenue leaks.
- Designed and deployed interactive Power BI dashboards to track substation load distribution, enabling senior engineers to identify and resolve **3** critical grid bottlenecks.
- Streamlined database query scripts to aggregate daily energy distribution logs, reducing reporting pipeline execution times by **30%**.

Data Analyst Intern

Thrillophilia

Apr 2025 – Jun 2025

Remote, India

- Analyzed **15K+** customer feedback and booking records using Python (Pandas) and SQL, identifying key package rating trends that optimized travel itinerary listings and increased booking rates.
- Developed operational Power BI dashboards to track regional sales performance and customer satisfaction metrics, saving **4+ hours** of manual compilation weekly.
- Queried relational databases using PostgreSQL to extract weekly operational metrics for business teams, reducing ad-hoc data request turnaround times by **25%**.
- Cleaned and validated raw booking datasets containing missing fields and duplicate entries, improving downstream data pipeline accuracy by **15%**.

PROJECTS

AI Competitive Intelligence Agent

LangGraph, GPT-4o, Next.js, Node.js (Node-Cron), ChromaDB, Docker

Jun 2026

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- Orchestrated an autonomous multi-agent system using LangGraph and GPT-4o, automating competitor tracking across **10+** sources and capturing **50+** daily data points.
- Deployed a self-updating RAG pipeline using ChromaDB and OpenAI Embeddings, improving semantic retrieval accuracy by **25%**.
- Integrated Tavily Search, Slack APIs, and **Node-Cron** to schedule and deliver automated briefings, reducing manual intelligence gathering by **15+ hours** weekly.
- Created a React/Next.js dashboard for real-time tracking, implementing an evaluation framework that achieved **94%** retrieval precision.

SEC EDGAR RAG & Analytics Pipeline

Python, FastAPI, Qdrant, Llama, Ragas, LangSmith, Docker, GitHub Actions

Jul 2026

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- Engineered an automated ingestion pipeline fetching SEC EDGAR filings, implementing section-aware chunking aligned with SEC Item boundaries to preserve semantic coherence and extract structured 10-K/10-Q sections.
- Developed a hybrid retrieval engine combining dense vector search (BAAI/bge-m3) and BM25 sparse retrieval via Reciprocal Rank Fusion (RRF), integrated with a BAAI/bge-reranker-v2 cross-encoder to optimize precision.

- Built a citation-grounded Llama generation pipeline with metadata-filtered Role-Based Access Control (RBAC), implementing guardrails for prompt injection, PII protection, and hallucination mitigation.
- Created an automated evaluation framework using Ragas integrated with CI/CD quality gates, deploying observability dashboards to track retrieval quality, token costs, and p50/p95 latencies.

SQL Data Warehouse & Analytics Pipeline

SQL Server, T-SQL, SSMS, Git

Jul 2026
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- Architected a 3-layer Medallion data warehouse in SQL Server, automating ETL pipelines to ingest and cleanse raw CRM/ERP data and reducing data latency by **40%**.
- Constructed star-schema dimensional models, writing **15+** optimized stored procedures, CTEs, and window functions in T-SQL.
- Optimized analytical SQL queries using indexing and query tuning, reducing query execution times by **40%** while maintaining version-controlled database scripts via Git.

EDUCATION

B.Tech in Computer Science & Engineering (AI & ML) — CGPA: 8.63/10

VIT Bhopal University

Aug 2023 – May 2027
Bhopal, India

Relevant Coursework: Data Structures & Algorithms, Database Management Systems (DBMS), Statistical Methods, Machine Learning, Data Mining, Python, Operating Systems, Cloud Computing

CERTIFICATIONS

- **AWS Certified Solutions Architect — Associate** — Cloud Services, Security, Infrastructure Deployment 2026
- **Data Science Bootcamp** (Krish Naik) — Python, Machine Learning, EDA, Feature Engineering 2025
- **Applied Machine Learning in Python** (University of Michigan) — Scikit-learn, Model Evaluation 2025
- **Cloud Computing** (NPTEL, IIT Kharagpur) — Cloud Architecture, GCP, Distributed Systems 2025